Student Use of Artificial Intelligence in Education

Abstract :

Artificial Intelligence (AI) has become increasingly relevant to student's academic activities, particularly in areas such as understanding difficult concepts, solving academic problems, preparing for examinations, summarising study materials, conducting research, coding, writing, and generating ideas. This research paper presents a descriptive analysis of a student survey containing 40 responses concerning the use of AI in education. The analysis is based exclusively on the data supplied in the student survey and its accompanying summary dashboard; no external sources have been used.

The findings indicate a high level of student engagement with AI tools. ChatGPT was the most widely used tool, reported by 36 respondents (90%), followed by Google Gemini with 30 respondents (75%). Seventeen respondents (42.5%) reported using AI several times a day. In relation to academic performance, 27 respondents (67.5%) reported either some improvement or significant improvement since using AI, while 11 respondents (27.5%) reported either some or no significant change in the negative direction. The major concerns identified by respondents were reduced critical thinking, lack of originality or creativity, and incorrect information. Despite these concerns, 35 respondents (87.5%) supported allowing AI in colleges in some form, mainly through clear guidelines or restrictions on particular academic activities.

Overall, the survey presents AI as a widely used academic resource that students generally perceive as useful, while simultaneously identifying concerns related to dependence, critical thinking, originality, accuracy, and responsible use. The findings suggest that the central issue emerging from the survey is not simply whether students use AI, but how they use it within the learning process.

Keywords: Artificial Intelligence, AI in Education, Students, Generative AI, Academic Performance, Critical Thinking, Student Behaviour, Educational Technology.

Introduction

Artificial Intelligence has become part of the academic activities of many students represented in this survey. Students reported using AI for a wide range of academic purposes, including understanding difficult concepts, solving academic problems, completing assignments, preparing for examinations, summarising study materials, conducting research, programming, improving written work, generating ideas, translation, and creating presentations.

The supplied survey was designed to examine several dimensions of student interaction with AI. These included the AI tools being used, frequency of use, time spent using AI for academic purposes, reasons for using AI, verification practices, academic performance, perceived changes in performance, concerns regarding AI, views on college policies, preferred forms of AI use, expectations regarding the future role of AI, and students' weekly study time outside class.

Rather than treating AI use simply as a technological behaviour, the survey provides information about how students perceive its educational value and its potential disadvantages. Respondents expressed both positive and negative views. Some described AI as useful for learning and understanding difficult topics, whereas others highlighted its potential to reduce critical thinking, originality, independent thinking, and the habit of consulting traditional learning resources.

This paper therefore analyses the survey from a descriptive research perspective and identifies the major patterns visible in the collected responses.

Objectives of the Study

The primary objective of this study is to analyse how students use Artificial Intelligence in education based on the supplied survey data.

The specific objectives are to:

- ***Examine the AI tools most commonly used by students.***
- ***Analyse the frequency of student AI usage.***
- ***Examine student's reported changes in academic performance after using AI.***
- ***Identify the major concerns students associate with AI.***
- ***Analyse student opinions regarding whether colleges should permit AI.***
- ***Examine student's preferred educational uses of AI.***
- ***Understand student's overall perceptions of AI in education.***
- ***Identify the relationship between the perceived benefits and concerns expressed by respondents.***

Research Methodologying

1. Research Design :

This study uses a descriptive survey research design. The objective is to describe patterns in student's reported use and perceptions of AI rather than to establish causal relationships.
The survey data were collected using structured questions covering demographic characteristics, academic background, AI usage behaviour, academic performance, concerns, institutional policy preferences, and attitudes toward AI in education.

2. Sample :

The supplied summary identifies 40 student responses. The survey includes respondents from different years of study and academic fields, including Science, Engineering/Technology, Arts & Humanities, Social Sciences, Management, Medicine/Healthcare, and other fields.

The field distribution represented in the supplied summary was:

Field of Study	Responses	Percentage
Science	18	45.00%
Engineering/Technology	11	27.50%
Arts & Humanities	3	7.50%
Social Sciences	3	7.50%
Other	3	7.50%
Management	1	2.50%
Medicine/Healthcare	1	2.50%
Total	**40**	**100%**

The dataset also records age group, gender, year of study and institution type for individual respondents.

3. Variables Examined :

The analysis considers the following major variables:

- *AI tools used*
- *AI usage frequency*
- *Academic hours spent using AI*
- *Reasons for AI use*
- *Frequency of verification*
- *Verification sources*
- *Current academic performance*
- *Change in academic performance since using AI*
- *Biggest concern about AI*
- *Opinion on college AI policies*
- *Preferred use of AI by students*
- *Perceived future importance of AI*
- *Weekly study hours outside class*
- *Overall opinion about AI in education*

4. Analytical Approach :

The study primarily uses descriptive analysis through frequencies and percentages. Open-ended student comments are also examined qualitatively to identify recurring themes.

Because the survey measures students' own perceptions, statements concerning academic improvement represent self-reported outcomes. They should therefore not be interpreted as proof that AI caused the reported academic changes.

Results

AI Tools Used by Students :

The survey shows a strong concentration around a small number of major AI platforms.

AI Tool	Students Reporting Use	Percentage of 40
ChatGPT	36	90.00%
Google Gemini	30	75.00%
Claude	10	25.00%
Perplexity	7	17.50%
NotebookLM	4	10.00%
GitHub Copilot	2	5.00%
Microsoft Copilot	2	5.00%

ChatGPT was reported by 36 respondents, making it the most commonly used AI tool in the survey. Google Gemini followed with 30 users. Claude, Perplexity, NotebookLM, GitHub Copilot, and Microsoft Copilot were used by smaller groups.

The results also indicate that several students use more than one AI platform. Individual responses include combinations such as ChatGPT with Gemini, Claude, Perplexity, NotebookLM, and coding-oriented tools. This suggests that AI usage among the respondents is not limited to a single platform.

Frequency of AI Usage

The frequency data demonstrate that AI is not merely an occasional resource for many respondents.

AI Usage Frequency	Responses	Percentage
Several times a day	17	42.50%
Several times a week	11	27.50%
Once a day	4	10.00%
Once a week	1	2.50%
Occasionally	6	15.00%
Rarely	1	2.50%
Total	**40**	**100%**

The largest group, 17 respondents, reported using AI several times a day. Another 11 respondents used AI several times a week. Therefore, the survey indicates that frequent AI use is a substantial part of the academic behaviour of the respondents. At the same time, the presence of occasional and rare users demonstrates that AI adoption is not uniform across all students.

Academic Purposes for Using AI

The responses show that students use AI for multiple academic purposes rather than for a single activity.

Commonly reported purposes included:

- ***Understanding difficult concepts***
- ***Solving academic problems***
- ***Completing assignments***
- ***Exam preparation***
- ***Summarising study materials***
- ***Research***
- ***Coding and programming***
- ***Writing or improving text***
- ***Generating ideas***
- ***Translation***
- ***Creating presentations***

The most repeatedly visible purpose in the individual responses is understanding difficult concepts. Several respondents also identified solving academic problems, examination preparation, research, and summarisation as important uses.

This pattern indicates that students are using AI across different stages of academic work, from initial understanding of a topic to completing or improving academic tasks.

Verification of AI-Generated Information

The survey also examined whether students verify the information provided by AI.

Individual responses included verification frequencies such as:

Always
Often
Sometimes
Rarely

The sources used for verification included textbooks, academic websites and journals, Google/search engines, official websites, teachers or lectures, and other AI tools. Some respondents also indicated that they usually do not verify AI-generated information.

This is an important finding within the survey because students simultaneously identified incorrect information as one of the leading concerns about AI.

The data therefore reveal a notable tension:
"Students recognise that AI information can sometimes be incorrect, while verification behaviour differs considerably between respondents."

This does not establish that students who verify information achieve better academic outcomes; the supplied data do not provide sufficient evidence to make such a causal or statistical conclusion.

Perceived Change in Academic Performance

One of the central questions in the survey concerned student's perceptions of academic performance since beginning to use AI.

Change in Academic Performance	Responses	Percentage
Some improvement	22	55.00%
Significant improvement	5	12.50%
No significant change	6	15.00%
Some decline	5	12.50%
Not sure	2	5.00%
Total	**40**	**100%**

A total of 27 respondents (67.5%) reported either some improvement or significant improvement in their academic performance.
At the same time, five respondents reported some decline, six reported no significant change, and two were uncertain.
The result should be interpreted carefully. The survey measures students' perceptions of change, not independently measured examination marks or grades before and after AI adoption.
Consequently, the data support the conclusion that many respondents perceived improvement, but they do not demonstrate that AI was the direct cause of that improvement.

Major Concerns About Artificial Intelligence

Students identified several concerns regarding the educational use of AI.

Biggest Concern	Responses	Percentage
Reduced critical thinking	10	25.00%
Lack of originality/creativit	8	20.00%
Incorrect information	8	20.00%
Overdependence on AI	4	10.00%
Privacy concerns	3	7.50%
No major concerns	3	7.50%
Other	3	7.50%
Academic dishonesty	1	2.50%
Total	**40**	**100%**

The most frequently selected concern was reduced critical thinking, identified by 10 respondents.

The next two major concerns were lack of originality/creativity and incorrect information, with eight respondents each.

These three categories alone account for 26 of the 40 responses, or 65% of the survey.

The individual comments reinforce these concerns. One respondent described AI as useful but warned that students may become dependent on it. Another stated that excessive dependence could reduce the ability of students to think creatively. Another respondent specifically mentioned reduced critical thinking and a reduced habit of referring to textbooks.

The qualitative responses therefore support the pattern visible in the quantitative data.

Student Attitudes Toward College AI Policies

The respondents were also asked whether colleges should allow AI.

Response	Students	Percentage
Yes, but with clear guidelines	23	57.50%
Only for certain academic	9	22.50%
Not sure	4	10.00%
Yes, without major restrictions	3	7.50%
No	1	2.50%
Total	**40**	**100%**

The data show that 23 respondents (57.5%) favoured allowing AI with clear guidelines.
Another 9 respondents (22.5%) supported allowing AI only for specific academic activities.
Taken together, 35 respondents, or 87.5%, selected one of these three forms of allowing AI:

- ***Yes, with clear guidelines***
- ***Only for certain academic activities***
- ***Yes, without major restrictions***

Only one respondent selected "No."
This indicates that the dominant position within this survey is not complete rejection of AI, but acceptance accompanied by some form of control, guidance, or defined boundaries.

Preferred Role of AI in Student Learning

The survey asked students to identify the best way for students to use AI.
Responses included:

- ***As a learning assistant***
- ***To explain difficult concepts***
- ***To improve existing work***
- ***As a source of ideas***
- ***To complete tasks quickly***
- ***For research assistance***
- ***Other uses***
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Among the individual responses, "as a learning assistant" and "to explain difficult concepts" appear repeatedly. Several respondents also selected AI as a source of ideas or as a tool for improving existing academic work.

This is consistent with several open-ended comments in which respondents presented AI as something that can support learning, particularly when students remain actively involved in the process.

At the same time, some students expressed concern that using AI primarily to complete tasks quickly could reduce the amount of independent thinking required from the student.

Perceived Future Importance of AI

The survey asked respondents whether AI would become essential in education in the future.
The available responses included:

- ***Definitely Yes***
- ***Probably Yes***
- ***Not Sure***
- ***Probably No***

The individual responses show that "Definitely Yes" and "Probably Yes" were frequently selected, although some students remained uncertain about AI's future role.

This suggests that many respondents regard AI as something that is likely to remain relevant to education rather than as a temporary academic tool.

However, the survey also makes clear that future acceptance does not necessarily mean unrestricted use. Many students simultaneously expressed concerns about overdependence, originality, critical thinking, and incorrect information.

Qualitative Analysis of Student Opinions

The open-ended responses add an important dimension to the numerical findings.

Several responses described AI positively. Students used expressions such as "very useful," "helpful for learning," "very useful," and "good." Other students described AI as particularly useful for understanding difficult topics.

However, positive opinions were frequently accompanied by qualifications.

One respondent described AI as a "double edged sword", explaining that its value depends on how effectively an individual uses it.

Another respondent stated that AI is a massive educational tool but warned that students may become too dependent on it.

Another respondent argued that AI can make academic assignments easier and faster, but this may occur instead of the student engaging in independent thinking.

A further response stated that limited AI use can be beneficial, while heavy reliance may weaken critical thinking and decision-making skills.

These comments reveal an important characteristic of the dataset: students generally do not describe AI in purely positive or purely negative terms. Instead, many responses present AI as a tool whose educational value depends on the manner in which it is used.

Discussion

The survey findings demonstrate three major characteristics of student AI use.

1. AI is becoming a regular academic tool among respondents:

A significant proportion of respondents reported frequent use of AI. The largest individual usage category was "several times a day," selected by 17 respondents. ChatGPT was used by 36 of the 40 respondents, while Google Gemini was used by 30. This means that, within this sample, AI is not confined to occasional experimentation. It is being integrated into everyday academic activities by many students.

2. Students perceive educational benefits :

The survey shows that 27 of the 40 respondents reported either some or significant improvement in academic performance following AI use.

Students also repeatedly reported using AI to understand difficult concepts, solve academic problems, prepare for examinations, conduct research, and summarise study materials.

The combination of these findings indicates that students perceive AI as having practical educational value.

However, this relationship should not be interpreted as causal. The survey does not establish whether AI directly improved students' academic performance.

3. Students are concerned about the consequences of excessive dependence :

The positive findings are accompanied by considerable concern.

Reduced critical thinking was the most frequently selected concern. Lack of originality/creativity and incorrect information were also major concerns. The qualitative responses repeatedly mention dependence, reduced thinking, and the possibility of completing academic tasks without sufficient independent engagement.

Thus, the survey presents AI use as a trade-off between academic assistance and independent intellectual engagement.

The data do not demonstrate that AI necessarily reduces critical thinking. Instead, they show that students themselves perceive this as a significant risk.

Implications of the Findings

The findings have several implications for understanding student AI behaviour.

First, the survey suggests that a complete prohibition of AI would not reflect the views of most respondents. The majority supported some form of institutional permission.

Second, the strongest support was for clear guidelines. Twenty-three respondents selected "Yes, but with clear guidelines," making this the largest policy category.

Third, the responses suggest that students distinguish between using AI as a support mechanism and allowing AI to replace their own academic work. The preferred uses reported in the survey frequently involve explanation, assistance, idea generation, and improvement rather than simply delegating all academic activity to AI.

Fourth, verification appears to be an important part of responsible AI use within the survey because incorrect information was one of the three most common concerns. However, verification practices varied considerably among respondents.

Proposed Interpretation of Responsible AI Use

Based only on the patterns in this survey, a useful interpretation of responsible student AI use can be developed around five principles:

1. Understand before depending :

Students frequently use AI to understand difficult concepts. This represents a learning-oriented use of AI, provided that the student remains involved in understanding the material.

2. Use AI as assistance rather than complete replacement :

Several respondents identified AI as a learning assistant. Other responses warned against allowing AI to replace independent thinking.

3. Verify important information :

Incorrect information was one of the most frequently identified concerns. The survey responses indicate that students use textbooks, academic websites, search engines, official websites, teachers, and other sources for verification.

4. Maintain originality :

Lack of originality and creativity was selected by eight respondents. This suggests that students recognise a potential problem when AI-generated material replaces individual ideas and expression.

5. Avoid excessive dependence :

Overdependence was explicitly selected as a concern by four respondents, while several open-ended comments also referred to excessive reliance on AI.

These principles are not external recommendations; they are an interpretation of the behavioural patterns and opinions expressed by respondents in the supplied survey.

Limitations of the Study

Several limitations should be considered when interpreting the findings.

1. Self-reported data :

The survey relies on students' own reports of AI usage and academic performance. Consequently, reported improvement cannot be treated as independently verified academic improvement.

2. Limited sample size :

The study is based on 40 responses. The findings therefore describe this particular sample and should not automatically be treated as representative of all college or university students.

3. Descriptive nature :

The survey provides descriptive information but does not establish causation. For example, the data cannot establish that AI caused an improvement or decline in academic performance.

4. Variation in individual usage :

Students use AI at different frequencies, for different purposes, and with different verification practices. Consequently, "AI use" represents a broad range of behaviours rather than one uniform activity.

5. Differences in survey responses :

Some respondents selected "Not Sure," "No Opinion," or "Prefer not to say" for particular questions. Such responses reflect uncertainty or non-disclosure and should not be treated as equivalent to either positive or negative opinions.

Conclusion

This study analysed a 40-response student survey concerning the use of Artificial Intelligence in education. The findings indicate that AI is already an important academic resource for many of the surveyed students.

ChatGPT was reported by 90% of respondents, while Google Gemini was reported by 75%. The largest group of respondents used AI several times a day. Students used AI across a broad range of academic activities, particularly for understanding difficult concepts, solving problems, preparing for examinations, summarising materials, conducting research, writing, coding, and generating ideas.

The survey also indicates a generally positive perception of AI's academic usefulness. A total of 67.5% of respondents reported some or significant improvement in academic performance since using AI. However, these results represent perceptions rather than independently measured academic outcomes.

At the same time, students identified substantial concerns. Reduced critical thinking was the most frequently selected concern, followed by lack of originality or creativity and incorrect information. These concerns were reinforced by open-ended responses describing the risks of excessive dependence and the possibility that students may rely on AI instead of engaging in independent thinking.

The policy findings are particularly notable. Although students expressed concerns about AI, 87.5% selected a response supporting some form of college permission for AI use. The largest group preferred AI to be allowed with clear guidelines.

Therefore, the central finding of this survey is that students do not appear to view AI simply as either beneficial or harmful. Their responses portray AI as a powerful academic tool whose value depends substantially on how it is incorporated into the learning process. The survey points toward a model in which AI can assist students while independent thinking, verification, originality, and personal academic responsibility remain important components of student work.

Data Source

This research paper uses only the student survey data supplied by the user, including the underlying survey responses and the corresponding summary information. No external publications, websites, studies, or secondary datasets have been used.

Reference

Supplied Student Survey Dataset. Project spread Sheet New one.pdf. Student survey on the use of Artificial Intelligence in education, 40-response dataset.